

Multiple Sequence Alignment of Hemoglobin Proteins

Description

This project involved performing multiple sequence alignment using Clustal Omega to compare hemoglobin protein sequences from different organisms.

Method

- Used Clustal Omega for multiple sequence alignment
- Input protein sequences from human, mouse, and zebrafish
- Analyzed conserved and variable regions

Results

The multiple sequence alignment showed a high level of similarity among hemoglobin proteins from human, mouse, and zebrafish.

Several fully conserved regions were identified (indicated by *), demonstrating identical amino acid sequences across all species.

Minor variations were observed at a few positions, reflecting evolutionary differences.

Clustal Omega Output

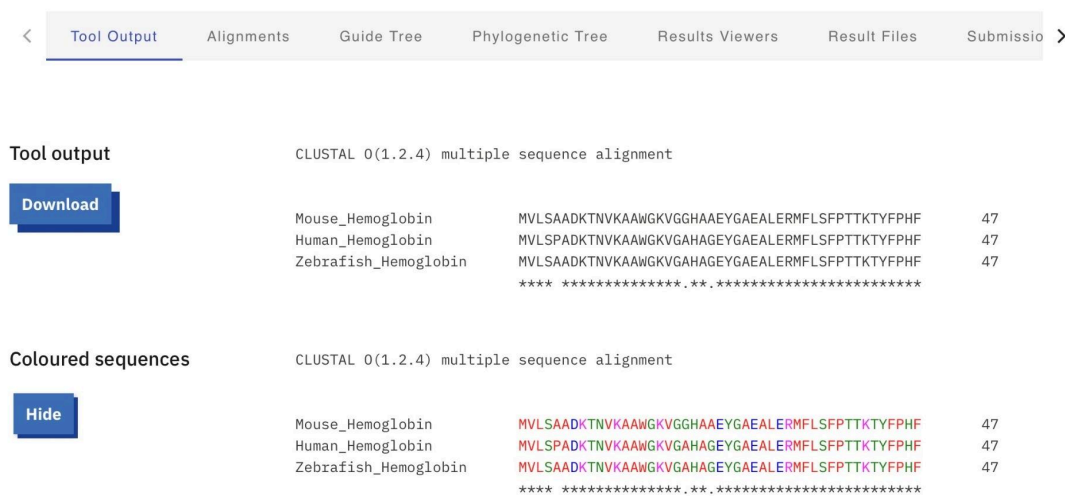


Figure 1: Multiple sequence alignment of hemoglobin proteins from human, mouse, and zebrafish, showing conserved regions (*) and minor variations.

Interpretation

The presence of highly conserved regions indicates that hemoglobin has a critical biological function that is preserved across different species.

The small variations observed suggest evolutionary adaptations while maintaining the core function of the protein.

Skills Demonstrated

- Multiple sequence alignment
- Comparative protein analysis
- Interpretation of conserved regions
- Use of bioinformatics tools (Clustal Omega)